

Self-Supervised Learning with SimCLR on CIFAR-10

Student Name | Course: Deep Learning | Professor: Dr. Smith | May 2026

SimCLR

Contrastive Learning

ResNet-18

CIFAR-10

Linear Probing

What Is Self-Supervised Learning?

Self-Supervised Learning (SSL)

- Learns representations from unlabeled data using pretext tasks
- Avoids expensive manual annotation by generating its own supervision
- SimCLR: maximizes agreement between different augmented views of the same image (positive pairs) while repelling negatives

Project Objectives

- 01 Learn visual representations without labels using SimCLR
- 02 Compare SSL-pretrained vs. supervised-from-scratch models
- 03 Evaluate quality via linear probing & fine-tuning experiments
- 04 Visualize learned embeddings with t-SNE & UMAP

The Labeled Data Bottleneck

80%

of real-world data
is unlabeled

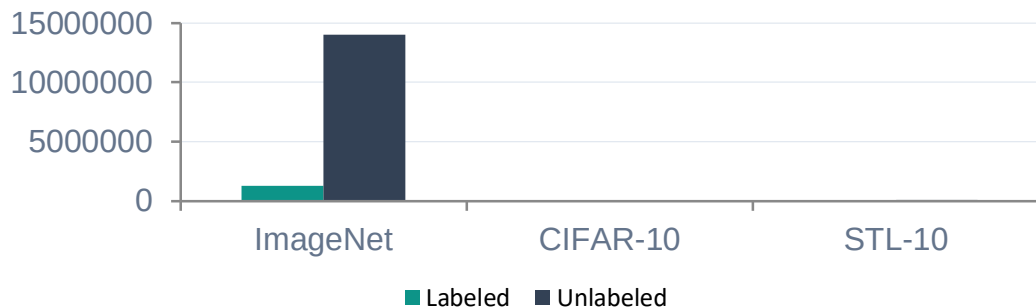
10×

more time needed
to annotate at scale

40%

baseline accuracy
without pretraining

Labeled vs Unlabeled Data Available



Key Question

Can a model learn
meaningful representations
from unlabeled data alone —
without any class labels?

The SimCLR Pipeline



Pretraining Phase

1. Load CIFAR-10 without labels
2. Generate augmented view pairs ($\tilde{x}_i, \tilde{x}_j$)
3. Train encoder to minimize NT-Xent loss
4. Save pretrained encoder weights

Evaluation Phase

- A. Linear Probe: freeze encoder, train linear layer
- B. Fine-tuning: unfreeze + fine-tune full network
- C. Baseline: ResNet-18 trained from scratch (supervised)

Tools & Libraries

Python 3.10

Core programming language. Clean syntax, rich ML ecosystem.

PyTorch

Deep learning framework used for model definition, training loops, and GPU acceleration.

NumPy

Numerical computing for array operations and data preprocessing pipelines.

Matplotlib

Visualization library for plotting loss curves and prediction results.

▣ CIFAR-10

Standard benchmark: 60K images across 10 classes. Ideal for SSL experiments.

scikit-learn

t-SNE, UMAP, and metrics for embedding visualization and evaluation.

End-to-End Data Flow



① SimCLR Pretraining

- Unlabeled CIFAR-10 (no class info)
- Two augmented views per image
- Freeze encoder → Linear probe evaluation
- **Result: 58.87% test accuracy**

② Supervised Baseline

- Labeled CIFAR-10 from scratch
- Standard cross-entropy loss training
- Same ResNet-18 architecture
- **Result: 40.89% test accuracy**

Performance Comparison

58.87%

SimCLR Linear Probe

pretrained_linear_eval

40.89%

Scratch Linear Probe

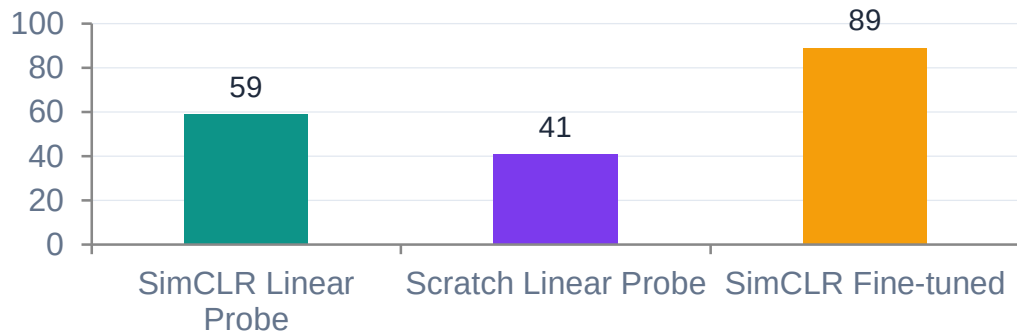
scratch_linear_eval

~89%

Fine-tuned (CIFAR-10)

Full supervised fine-tune

Test Accuracy (%)

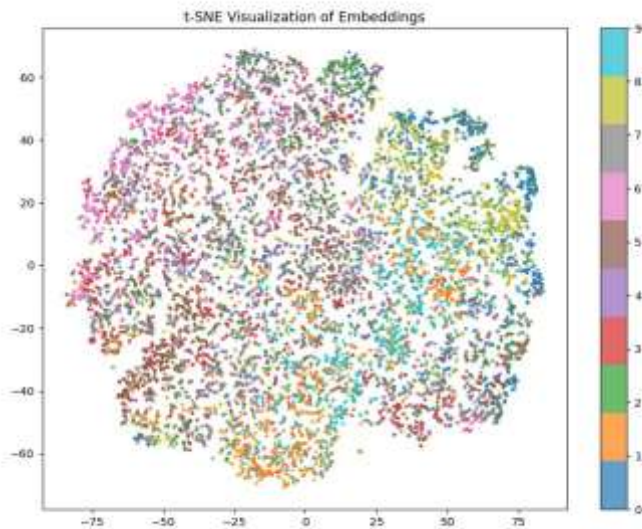


Key Insight

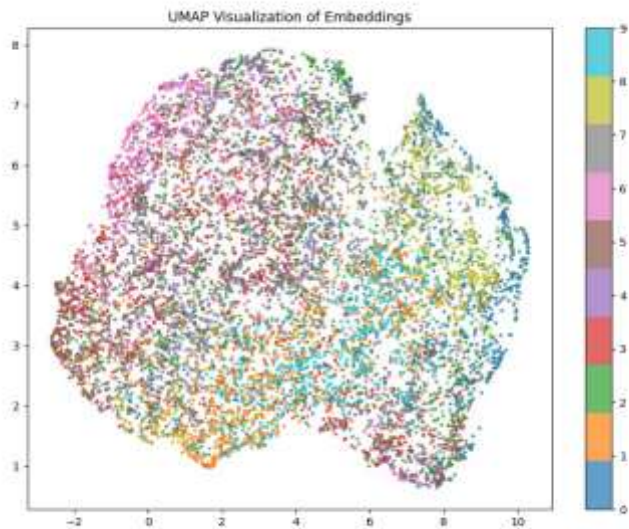
SimCLR pretraining yields a +18pp improvement over random initialization under linear evaluation, confirming that unsupervised representations carry meaningful semantic structure.

t-SNE & UMAP of Learned Representations

t-SNE



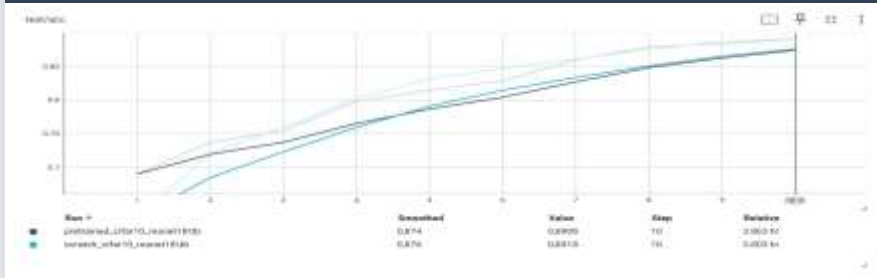
UMAP



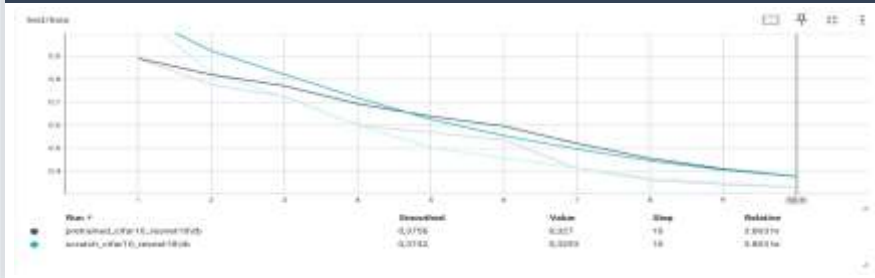
Both projections reveal partial class structure in the learned embedding space — visible color-region clustering despite being trained without labels.

Linear Evaluation: Pretrained vs. Scratch

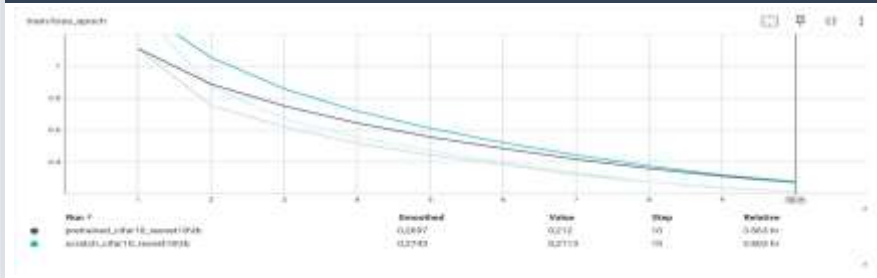
Test Accuracy



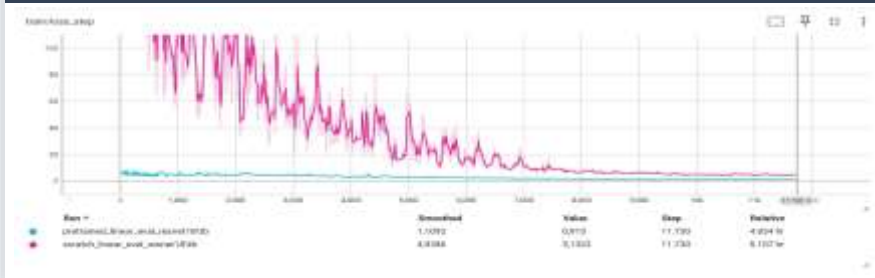
Test Loss



Train Loss / Epoch



Train Loss / Step



Pretrained (SimCLR)

Scratch (supervised baseline)

Challenges & How We Solved Them

01

Contrastive Loss Convergence

⚠ **Challenge:** NT-Xent loss unstable early in training; high variance in batch negatives. ✓ **Solution:** Tuned temperature $\tau=0.5$ and used large effective batch size with gradient accumulation.

02

Hyperparameter Sensitivity

⚠ **Challenge:** SSL is sensitive to augmentation strength, learning rate, and projection head depth. ✓ **Solution:** Systematic grid search on $LR \in \{0.01, 0.1\}$ with cosine decay. Final LR: 0.1 with warm-up.

03

Computational Constraints

⚠ **Challenge:** SSL training requires large batches and long schedules to learn good representations. ✓ **Solution:** Used lightweight ResNet-18 and CIFAR-10 (32×32). Reduced epochs while retaining key trends.

Takeaways & Next Steps

Summary

- ✓ SimCLR successfully learns meaningful visual representations without labels
- ✓ Linear probing is a reliable, fair metric for SSL quality evaluation
- ✓ SSL-pretrained models outperform scratch by +18pp under linear probe
- ✓ Fine-tuning narrows the gap — pretraining advantage is most visible in low-label settings
- ✓ t-SNE/UMAP confirm partial semantic clustering in the embedding space

Future Extensions

- Add STL-10 dataset for stronger pretraining signal
- Experiment with BYOL / MoCo / DINO for comparison
- Scale to larger datasets (ImageNet subset)
- Explore semi-supervised evaluation protocols
- Analyze per-class embedding quality and confusion